

Reg. No.: 29BEC1127

Name :



VIT

Vellore Institute of Technology

(Approved as the University under section 3 of UGC Act, 1956)

Continuous Assessment Test II – March 2024

Programme	: B.Tech (ECE)	Semester	: WS 2023-24
Course	: Electronic Materials and devices	Code	: BECE201L
		Class Nbr	: CH2023240501111 CH2023240501112
Faculty	: Dr. Sunil Kumar Pradhan Dr A Arivarasi	Slot	: D1+TD1
Time	: 90 Minutes	Max. Marks	: 50

Answer ALL the questions

Q.No.	Sub. Sec.	Questions	Marks
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Let's consider a silicon sample with unit cross sectional area as shown in the Figure 1, which is in thermal equilibrium. The following information is given:

$T = 300$ K, Electronic charge $= 1.6 \times 10^{-19}$ C, Thermal voltage $= 26$ mV

Electron mobility $= 1350$ cm²/V-s

(a) Find the magnitude of the electric field at $x = 0.5 \mu\text{m}$ [3 marks]

(b) Find the magnitude of the electron drift current density at $x = 0.5 \mu\text{m}$ [3 marks]

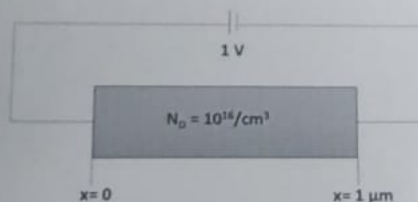


Figure 1

- (c) Let's assume electronic charge (q) $= 1.6 \times 10^{-19}$ C, $kT/q = 25$ mV and electron mobility (μ_n) $= 1000$ cm²/V-s. If the concentration gradient of electrons injected into a p-type silicon sample is 1×10^{21} cm⁻⁴, then find the magnitude of the electron diffusion current density (in A/cm²). k = Boltzmann constant,

T = absolute degree temperature

[4 marks]

- (a) Analyse, narrate, and provide equivalent circuits of two different mechanisms by which a PN junction breaks in a diode. [5 marks]

- (b) Differentiate "Normal" and "Breakdown diode" in terms of I-V Characteristics.

[5 marks]

Let's consider a bar of silicon, which is doped with boron concentration of 10^{16} cm^{-3} and it's assumed to be fully ionized. It is exposed to light as shown in the Figure 2, such that the electron-hole pairs are generated throughout the volume of the bar at the rate of

$10^{20} \text{ cm}^{-3}\text{s}^{-1}$. If the recombination lifetime is $100 \text{ }\mu\text{s}$, intrinsic carrier concentration of silicon is 10^{10} cm^{-3} and assuming 100% ionization of boron, then find the approximate product of steady-state electron and hole concentration due to this light exposure.

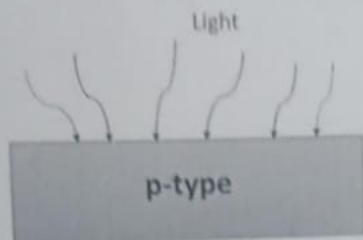


Figure 2

Referring to the Figure 3, the switch is in position 1 initially and steady state condition exist from $t=0$ to $t=t_0$, the switch is suddenly thrown into position 2. Find the current (i) through the $10\text{k}\Omega$ resistor as a function of time (t), from $t=0$. Also sketch (i vs t) showing the magnitudes of the current at $t=0$, $t=t_0$ and $t=\infty$.

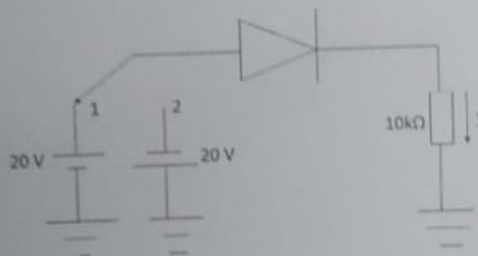


Figure 3

Find the reason, the purpose of impedance matching in transistor coupled circuits. For example, using microphone or radio, in order to amplify the signal, how impedance matching helps? Comprehend & narrate about the kind of transistor configuration used for effective impedance matching in amplifiers.